

Eco-Friendly Automatic Devices for the Environment

Innovations in Trial & International Markets

Satyam Maury

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Subject

Environmental Studies (EVS)



Why We Need Eco-Friendly Automatic Devices

Eco-friendly automatic devices use **IoT, AI, and robotics** to monitor, clean, and optimize natural systems — reducing human error and scaling impact.

Water & Oceans

8 million tons of plastic enter oceans yearly

Air Pollution

99% of the global population breathes unsafe air

Waste Crisis

Only 9% of global plastic is actually recycled

Energy Waste

Buildings consume 40% of global energy

- 📝 **Speaker Notes:** Let's start with the why. Our planet faces four critical environmental crises: ocean plastic pollution, toxic air, inefficient waste management, and massive energy waste. Manual solutions simply cannot scale to meet these challenges. Eco-friendly automatic devices — powered by IoT sensors, AI algorithms, and robotics — can operate continuously, make intelligent decisions, and cover areas humans cannot. As CSE students, understanding these systems is directly relevant to our field.

Automated Devices Cleaning Our Waters



Seabin Project 🌊

Autonomous floating bin — collects microplastics, oil, and debris in marinas. Pumps 25,000 L/hour.

Mr. Trash Wheel 🚗

Solar + hydro-powered interceptor in Baltimore. Removed **2,000+ tons** of trash from waterways.

WasteShark 🦈

Autonomous drone skimming surface pollutants. GPS-mapped, collects 160 kg/day.



Speaker Notes: Moving to water and ocean cleaning. The Seabin Project is like an automated trash can for harbors — it runs 24/7, pumping water through a catch bag. Mr. Trash Wheel in Baltimore uses solar and water current energy to intercept trash before it reaches the ocean — it has removed over 2,000 tons. The WasteShark is an autonomous drone that skims the water surface, collecting microplastics and pollutants. All three use sensors and automation to operate without constant human intervention.

Smart Devices Breathing Life Back into Cities



Smog Free Tower

Rotterdam's 7m tower ionizes PM2.5/PM10 particles. Cleans **30,000 m³/hour** — captured particles compressed into jewellery.



CityTree

Automated moss biofilter with IoT sensors. Equivalent to **275 trees**. Monitors CO₂, NO_x, and particulates in real time.



Automated Moss Filters

Indoor living walls with automated irrigation and air-flow systems. Reduce VOCs and improve cognitive performance by up to **15%**.

- ❏ **Speaker Notes:** For air quality, the Smog Free Tower in Rotterdam uses ionization technology to capture fine particulate matter — it literally turns smog into jewellery. CityTree is a brilliant urban solution: a moss-based biofilter with embedded IoT sensors that monitors air quality in real time and is equivalent to 275 trees in a fraction of the space. Indoor automated moss filters are also gaining traction in smart buildings, reducing volatile organic compounds and even improving worker productivity.

AI & IoT Transforming Waste Sorting



AMP Robotics

AI-powered robotic arms sort **160 items/min** using computer vision. Accuracy: 99%. Trained on millions of waste images.

Solar Compactors (BigBelly)

Solar-powered compactors with IoT fill-level sensors. Reduce collection trips by **80%**, cutting fuel emissions.

Smart Bins (Bin-e)

Automatic waste recognition, compression, and sorting. Cloud-connected for municipal waste analytics dashboards.



Speaker Notes: Smart waste management is where AI and robotics shine. AMP Robotics uses deep learning and computer vision — the robot identifies waste type in milliseconds and sorts 160 items per minute with 99% accuracy. BigBelly solar compactors use IoT sensors to signal when they're full, dramatically reducing unnecessary collection truck trips. Bin-e is a fully automated smart bin that recognizes, compresses, and sorts waste at the point of disposal. These are direct applications of the AI and ML concepts we study in CSE.

Automating Efficiency Across Energy & Agriculture



Smart Thermostats (Nest, Ecobee)

Learn occupancy patterns, auto-adjust HVAC. Save **10–15%** on energy bills. Integrate with renewable energy grids.



Automated Smart Irrigation

Soil moisture + weather API-driven drip systems. Reduce water waste by **50%**. Deployed across India's PMKSY scheme.



Smart Energy Grids

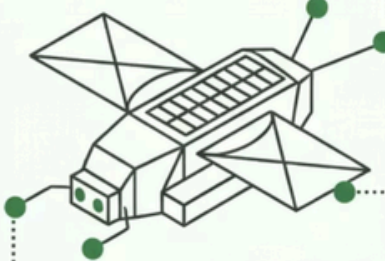
AI-driven load balancing with solar/wind integration. Predict demand, reduce transmission losses by **8–12%**.

📌 **Speaker Notes:** Energy and resource optimization is critical for sustainability. Smart thermostats like Nest use machine learning to learn your habits and automatically adjust heating and cooling — saving 10 to 15 percent on energy. Smart irrigation systems use soil moisture sensors combined with weather APIs to deliver water only when and where needed, cutting agricultural water waste by up to 50 percent. India's PMKSY scheme is actively deploying these. Smart energy grids use AI to balance renewable energy supply with demand in real time.

Cutting-Edge Devices Being Tested Globally

**ROBOBEES
(HARVARD)**

**TINY
AUTONOMOUS
DRONES**



SOLAR-POWERED

**MICRO-SENSORS
FOR NAVIGATION**

(HARVARD)

Replace declining
honeybees, pollinate crops

**THE OCEAN CLEANUP
SYSTEM 03**

**AUTONOMOUS
PLASTIC-EATING
BOAT**



AI VISION

**TARGETS
GREAT
PACIFIC
GARBAGE
PATCH**

**THE OCEAN
CLEANUP
SYSTEM 03**

Collects 90% of ocean
plastic by 2040



RoboBees — Harvard University

Tiny autonomous drones mimicking bee pollination. Solar-powered with micro-sensors. Addresses global honeybee decline threatening **75% of food crops**.



Ocean Cleanup System 03

Autonomous AI-guided vessel targeting the Great Pacific Garbage Patch. Uses computer vision to identify and collect plastic. Goal: remove **90% of ocean plastic by 2040**.



Both devices are in active field trials — representing the frontier of environmental automation.



Speaker Notes: Now let's look at what's on the cutting edge. Harvard's RoboBees are tiny autonomous drones designed to pollinate crops — this addresses the critical decline of honeybee populations which threatens 75 percent of our food crops. The Ocean Cleanup System 03 is an autonomous AI-guided vessel that uses computer vision to identify and collect plastic from the Great Pacific Garbage Patch. The goal is to remove 90 percent of ocean plastic by 2040. Both are in active field trials right now.

How AI, ML & IoT Power These Devices

01

Sensing Layer — IoT

Sensors (PM2.5, moisture, GPS, cameras) collect real-time environmental data streams.

02

Processing Layer — Edge/Cloud

Embedded microcontrollers + cloud platforms (AWS IoT, Azure) process and filter sensor data.

03

Intelligence Layer — AI/ML

Computer vision, neural networks, and predictive models enable autonomous decision-making.

04

Action Layer — Automation

Robotic actuators, motors, and valves execute physical environmental interventions.



Speaker Notes: This slide is especially relevant for us as CSE students. Every eco-friendly automatic device follows a four-layer architecture. First, IoT sensors collect raw environmental data. Second, edge devices or cloud platforms process that data. Third, AI and machine learning models — like convolutional neural networks for image recognition — make intelligent decisions. Fourth, actuators and robotic components physically act on those decisions. This is essentially the same pipeline we study in our AI and IoT courses, applied to real-world environmental problems.

The Flip Side: Barriers to Adoption

High Initial Cost

IoT devices, AI chips, and robotics require significant capital.
Limits deployment in developing regions.

E-Waste Paradox

Devices themselves become electronic waste. Lifecycle management and recyclable design are critical.

Maintenance & Reliability

Harsh outdoor environments degrade sensors and actuators.
Requires skilled technicians and spare parts.

Data Privacy & Security

IoT networks are vulnerable to cyberattacks. Environmental data must be secured and ethically managed.

📄 **Speaker Notes:** No technology is without challenges. High initial costs limit deployment in developing countries where environmental need is greatest. There's also an ironic e-waste paradox — devices built to help the environment can become electronic waste themselves if not designed for end-of-life recycling. Maintenance in harsh outdoor conditions is technically demanding. And finally, IoT networks collecting environmental data raise cybersecurity and data privacy concerns that we as engineers must address proactively.

The Road Ahead: Technology as an Environmental Ally

Key Takeaway

Automatic eco-devices are not a luxury — they are essential infrastructure for a sustainable future.

Future Scope

Swarm robotics, AI-driven carbon capture, and biohybrid devices will define the next decade.

Our Role as CSE Engineers

Build smarter algorithms, design secure IoT systems, and advocate for green tech policies.

References

- Seabin Project — seabinproject.com
- AMP Robotics — amrobotics.com
- The Ocean Cleanup — theoceancleanup.com
- WHO Air Quality Guidelines, 2021
- Harvard RoboBees Research — wyss.harvard.edu

 **Speaker Notes:** To conclude — eco-friendly automatic devices represent one of the most impactful applications of Computer Science in the real world. From cleaning our oceans to purifying our air, from sorting waste to optimizing energy, technology is becoming an essential environmental ally. The future holds even more promise: swarm robotics, AI-driven carbon capture, and biohybrid devices. As CSE engineers, we have both the opportunity and the responsibility to build these systems. Thank you for your attention — I'm happy to take any questions.